GSWA HyLogger Capstone

Industry Adoption Stream (P4) – Preliminary Data Analysis

Analysis Basis

The analysis in this document is based on one hole: 05GJD001.

05GJD001 is a real drill hole downloaded from NVCL and analyzed end-to-end (spectra, TSA mineral results, tray photos) as a single worked example. The brief's step 1 asks for 3-5 holes before drawing conclusions - findings here should be treated as a proof-of-concept on one hole, not yet validated across the dataset.

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1. Data Sources at a Glance

Source	Purpose
NVCL	Open HyLogger data - TSG spectral files + core tray photos, one zip per hole.
WAMEX	Exploration reports by A-number - collar location, depth logging, assay data.
WAPIMS	Oil & gas well data, searched by Well ID.
GeoView	GIS/spatial viewer - regional context, spatial search.
pyTSG	Python package to read TSG spectral files.
Vocabularies	GSWA dictionary of HyLogger/TSG technical terms (in progress).

1.1 Which Sources Matter Most for Industry Adoption

High priority

- NVCL **-** the primary source itself: the HyLogger spectra and TSA mineral summaries. Every other source only becomes useful once NVCL data is already in hand.
- WAMEX - the only source for the hole's spatial location (lat/long), lithology logging, and assay/geochemistry data.
- pyTSG - not a data source. Use to read spectral/TSA files (.tsg/.bip) files.

Medium priority

- GeoView - spatial context and search; useful for the map component of the front-end.
- Vocabularies - decodes cryptic codes (uTSAS, CHLORITE, DARK-MICA, etc.) but still incomplete per Julian.

Low priority - likely skip

WAPIMS - oil & gas wells, not mineral exploration drill cores; out of scope for this brief.

1.2 What's Inside a WAMEX Report, in Simple Terms

- Collar location - the exact spot on the ground where the drill hole starts (like a pin on a map), plus which direction and angle it was drilled at. Every depth in the hole is measured starting from this one point.
- Depth logging - the geologist's notes on what type of rock was found at each depth interval - the closest thing to the brief's 'expert logs' needed for evaluation.
- Assay data - lab test results showing how much of something valuable (like gold) was found at each depth interval - the economic answer behind 'ore vs waste'.

2. Understanding Drilling & Core Data Concepts (Q&A)

Basics

Q: What is a drill hole?

A: The entire borehole at one location, from the surface (0 m) down to total depth.

Q: What is a drill run?

A: One cycle of the core barrel: drill until full (or driller stops), pull up, extract core, redeploy.

Q: Is a drill run the same as one barrel?

A: Yes - one run = one barrel trip. Same reusable barrel each time, not a new one per run.

Q: Does one hole contain many runs/barrels?

A: Yes - a deep hole (500 m+) can have 100+ runs.

Depth

Q: How is a hole's start/end depth decided?

A: Start = always 0 m (the collar). End (total depth) = decided by the geologist, based on target depth, real-time drilling results, or budget/technical limits.

Q: Why does the example hole's data start at 150 m, not 0 m?

A: HyLogger scanning likely only began at 150 m (shallow weathered rock is often skipped), not that drilling itself started there. Confirmed with real data: the first logged sample's Depth = 150.000 m exactly.

Trays

Q: What is a core tray?

A: The wooden box storing recovered core from a run; numbered sequentially per hole (0001, 0002…).

Q: Can one hole have many trays?

A: Yes - e.g. a 300 m hole could need 60+ trays. 05GJD001 has 33 trays.

Q: Are tray depth boundaries planned in advance?

A: No - a new tray just starts once the current one is full. Not geologically meaningful.

Rows

Q: What is a row within a tray?

A: A ~1 m channel inside the tray. Core fills row 1 left-to-right, then wraps into row 2, etc.

Q: Is a row the same as one drill run?

A: Not necessarily. Rows are fixed-length containers; runs vary in length. A run can span several rows, or several runs can sit inside one row.

Q: If a run is longer than one row, what happens?

A: It just continues into the next row - no marker at the wrap point, since it's the same run.

Wooden Blocks

Q: What is a wooden block?

A: A small divider placed in a row marking a depth boundary - labelled S (start), E (end), or CL (core loss/cavity).

Q: Does a wooden block mark the end of a run/barrel?

A: Usually yes (S/E pair) - but it can also be a CL marker for a core-loss interval, not a real run boundary. Check the label to tell which.

Q: How is the tray filled - doesn't it break between runs?

A: Core is laid by hand in sequence. Rock often breaks naturally (fracturing) - unrelated to run boundaries. No physical gap is needed at a run's end: a wooden block is simply inserted, and the next run's core continues right after it.

Naming

Q: Is 05GJD001 the hole number?

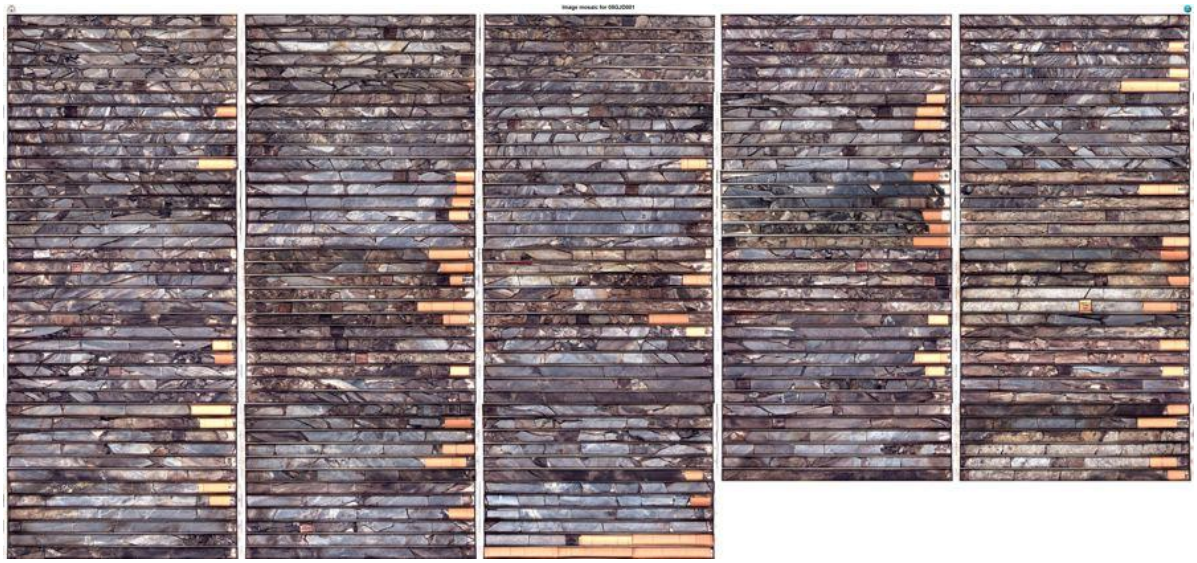
A: Yes - the drill hole ID, used to tie together tray photos, NVCL filenames, and (via cross-reference) the WAMEX A-number.

3. Visual Reference - Tray Photo & Mosaic

Figure 1 - Tray 0001 (150.0-154.7 m), one of 33 trays for hole 05GJD001. Six rows, read left to right, top to bottom = increasing depth. Handwritten wooden-block depth tags are visible.



Figure 2 - Full-hole mosaic: all 33 trays for 05GJD001 stitched into one continuous image.



4. The HyLogger File Folder - What's Inside

One hole's NVCL download extracts to a folder containing the tray photos above plus a set of spectral data files. Using 05GJD001 as the worked example:

File	What it is	How to view it	Related files
05GJD001_0001.jpg ... _0033.jpg	Tray photos (33 total)	Image viewer	Feeds into _Mosaic.jpg
05GJD001_tsg_Mosaic.jpg	All 33 trays stitched together	Image viewer	Built from the 33 tray jpgs
05GJD001_tsg.tsg	Index file for the NIR/SWIR scan	Python (pyTSG) - see Section 7	Paired with 05GJD001_tsg.bip
05GJD001_tsg.bip	NIR/SWIR reflectance numbers + TSA mineral summaries (scalars)	Python (pyTSG)	Paired with .tsg; corrected using .cal. Section 7.6: scalars = the brief's recommended starting dataset (TSA mineral summaries).
05GJD001_tsg.cal / .ini (+ _1/_2/_3.ini)	Calibration / instrument settings	Not opened directly - used automatically	Belongs to the NIR/SWIR set
05GJD001_tsg_tir.tsg	Index file for the TIR scan	Python (pyTSG)	Pairs with tsg_tir.bip
05GJD001_tsg_tir.bip	TIR reflectance numbers	Python (pyTSG)	Paired with tsg_tir.tsg; corrected using tsg_tir.cal. Background data per Section 6, not the brief's recommended starting point.
05GJD001_tsg_tir.cal / .ini	Calibration / settings for TIR	Not opened directly	Belongs to the TIR set
05GJD001_tsg_cras.bip	High-res core image, possibly pre-classified	Python (pyTSG) - currently fails, see Section 7.7	Standalone; largest file (373 MB)
05GJD001_tsg_hires.dat	High-resolution scan, one value per depth sample	Python (pyTSG)	Standalone; not spectral

5. NIR/SWIR vs TIR - Two Different Measurements

Note: the brief recommends starting from TSA mineral summaries (Section 7.6), not raw spectra.

The electromagnetic spectrum, in order of wavelength

Visible light	~400-700 nm	(what the human eye sees)
NIR	~700-1000 nm	(Near-InfraRed)
SWIR	~1000-2500 nm	(ShortWave InfraRed)
TIR	~8000-14000 nm	(Thermal InfraRed, aka LWIR)

Confirmed from 05GJD001's real data (Section 7): the 'NIR/SWIR' file actually spans 380-2500 nm (visible light through NIR through SWIR in one scan), while the TIR file spans 6000-14500 nm - a separate, much longer wavelength region.

Why they measure different things

- NIR/SWIR: shines light on the rock and measures what bounces back (reflectance). Spectral dips come from OH (hydroxyl), H2O, and CO3 (carbonate) bonds - excellent for clays, micas, chlorites, and carbonates.
- TIR: measures the rock's own thermal emission. Its features come from Si-O bonds in silicates - much better for anhydrous minerals like quartz and feldspar, which SWIR barely sees.

Why it matters for the project

This explains why SWIR and TIR can show different results at the same depth (raised in the meeting).

Real minerals identified in 05GJD001's SWIR data - Chlorite, Siderite, Phlogopite, Biotite - are all OH-bearing or carbonate minerals.

Quartz or feldspar at the same depth would likely only show up in TIR - an open question for whether the MVP's alteration/ore-waste classifier needs TIR at all.

6. Reading the Data with Python

Everything below was pulled directly from hole 05GJD001 using GSWA's pyTSG.

6.1 Loading a hole

- Install: pip install pytsg
- Extract the hole's NVCL zip so the .tsg/.bip/.cal/.ini/.dat files sit directly inside one folder.
- Call read_package(folder) - it scans the folder by filename suffix and automatically pairs and loads the matching files.

```
from pytsg import parse_tsg

data = parse_tsg.read_package(folder, read_cras_file=False)

data.nir.spectra          # shape (25938, 531) - the reflectance numbers

data.nir.sampleheaders    # depth (D), tray (T), row (L), etc. per sample

data.nir.scalars          # 151 columns incl. Min1/Wt1/Grp1 mineral results
(TSA summaries)
```

6.2 What comes back: .nir and .tir

Two separate buckets of results, one per sensor.
.nir holds the NIR/SWIR data (380-2500 nm);
.tir holds the TIR data (6000-14500 nm).

They're two physically different instruments that scanned the same core, so pyTSG keeps their results separate rather than mixing them together.

6.3 Understanding the shape: 25,938 x 531

- 25,938 = how many points along this particular hole were measured. Picture walking the full length of core (all 33 trays end to end) and stopping every ~8mm to take a reading - that gives 25,938 stops. Each row of .spectra is one stop.
- 531 = how many individual wavelengths are measured at each stop. A normal camera records 3 numbers per pixel (red, green, blue). This sensor records 531 very fine wavelength slices across 380-2500 nm at every stop - 531 narrow 'colors' instead of 3 broad ones. That's what makes it hyperspectral.
- Put together: 25,938 stops x 531 wavelength readings per stop = the full .spectra array. .tir uses the same 25,938 stops but only 341 wavelength slices, since its sensor divides its range differently.

Note: 25,938 is specific to 05GJD001, not a universal number - it depends on how much core this hole had (33 trays) and the sample spacing (~8mm). A deeper or shallower hole would have a different row count. 531 (the wavelength bands), however, is fixed by the instrument and stays the same across every hole. When writing code for multiple holes, read the shape dynamically (data.nir.spectra.shape[0]) rather than hard-coding 25938.

6.4 Real example: depth and position (sampleheaders)

What sampleheaders actually is

When a hole's spectral data loads, it comes with 25,938 spectra - but a spectrum alone is just a wall of numbers. sampleheaders is a separate table that says where each spectrum came from. Think of it like a caption for every photo in an album: the spectrum is the photo, sampleheaders is the caption.

Walk through one real row

Row 0 of sampleheaders for 05GJD001 looks like this:

```
D = 150.000008    X = 6.263977    T = 0001    L = 1    H = 05GJD001
```

In plain English, this one row says: 'This particular spectrum was measured at 150.000 m depth, 6.26 mm along the tray, inside Tray 0001, Row 1, from hole 05GJD001.' That's the whole idea - six columns, six facts about where that one measurement came from.

What each letter stands for, and how we know

The letters (D, X, T, L, H, P) are just what pyTSG's underlying file format calls them - cryptic on purpose, since it's an internal engineering format, not something designed for readability. These meanings were not guessed - they were confirmed by matching the real printed numbers against facts already known from the tray photo (Section 4):

Column	Confirmed meaning	Evidence
D	Depth (m)	First 5 values ~150.000 m, matching Tray 0001's known start depth
X	Distance along section (mm)	Values 6.26, 14.26, 22.26... increasing in 8mm steps
T	Tray number	"0001" for these rows, matching Tray 0001
L	Line/row within tray	1 for early rows, consistent with row 1
H	Hole ID	Literally "05GJD001"
P	Position/sample index	Matches sample + 1

Why this table matters

Without it, there would be 25,938 spectra with no way to know which depth each one belongs to. sampleheaders is what lets you say 'give me the spectrum at 152 m' instead of just 'give me spectrum #4127' - it is the link between the raw numbers and something geologically meaningful.

6.5 How the 25,938 Spectra Map to sampleheaders

The mechanism: position, not a shared ID

There is no explicit 'key' column linking spectra and sampleheaders. Instead, they are two parallel tables of the exact same length (25,938 rows), built and kept in the same order. The link is purely: row i in one = row i in the other, because that is how pyTSG constructed them - not because of any ID column matching them up.

Like two spreadsheets sitting side by side:

Row #	spectra (531 numbers each)	sampleheaders
0	[0.274, 0.273, 0.271, ...]	D=150.000, T=0001, L=1
1	[0.278, 0.276, 0.274, ...]	D=150.007, T=0001, L=1
2	[0.282, 0.279, 0.277, ...]	D=150.014, T=0001, L=1

Row 0 of spectra belongs with row 0 of sampleheaders simply because they are both row 0 - nothing more sophisticated than that.

Using it in practice - two directions

Spectrum to depth (have an index, want to know where it's from):

```
data.nir.spectra[100]              # the 531-number spectrum at position 100

data.nir.sampleheaders.iloc[100]   # its depth, tray, row - the same position
```

Depth to spectrum (want a specific depth, need to find its spectrum) - the pattern actually used most often, since nobody thinks in row-index numbers, they think in depth:

```
idx = (data.nir.sampleheaders["D"] - 152.0).abs().idxmin()

spectrum_at_152m = data.nir.spectra[idx]
```

One thing to be careful of

If spectra or sampleheaders are ever filtered, sorted, or have rows dropped independently of each other, this alignment breaks immediately - it only works because both arrays are the same length and have not been reordered relative to each other. Always operate on both together, or use the row index consistently across both.

6.6 Real example: mineral results (scalars) - the brief's recommended starting dataset

This is the TSA mineral summary data the brief explicitly says to start from.

What TSA Is

TSA stands for The Spectral Assistant, the automated mineral-identification module built into CSIRO/GSWA's TSG (The Spectral Geologist) software. It is the 'machine's best guess at which minerals sit at each depth', per the brief's own wording - it looks at the raw reflectance spectrum at each depth and automatically assigns a mineral name, confidence weight, and mineral group, without a human reviewing each result.

In the actual data files, TSA's output is what populates the scalars table (column names ending in TSAS, e.g. Min1 uTSAS, Wt1 uTSAS, Grp1 uTSAS) discussed below. This is why the brief recommends starting here: TSA has already done a first-pass mineral classification for every hole in NVCL, so the team does not need to build one from raw spectra to get moving.

Real numbers from 05GJD001 (this hole only - see Analysis Basis at the top of this document):

151 columns of interpreted results per sample. Of 25,938 samples, 16,006 (62%) have an actual identified mineral; the rest carry a 'no data' sentinel value (-3.402823e+38) that should be filtered out, not treated as zero - this gap needs an explicit handling decision in the ETL step.

Depth (m)	Min1 uTSAS	Wt1 uTSAS	Grp1 uTSAS	Error
150.014	Chlorite-FeMg	0.669	CHLORITE	91.0
150.028	Chlorite-FeMg	0.701	CHLORITE	160.2
150.035	Siderite	0.528	CARBONATE	149.6
150.063	Phlogopite	0.755	DARK-MICA	166.6
150.070	Biotite	0.532	DARK-MICA	95.4

Min1 = top-identified mineral name;
Wt1 = proportion/confidence weight (0-1);
Grp1 = broader mineral group; Error = uncertainty metric.

This is a real, working mineral classification already computed by GSWA/CSIRO's own software, per depth, for this hole. Important: it is an automated machine output, not a human-verified expert log -

Why 151 columns

- ~75 columns: up to 25 detected spectral absorption features per sample, each described by Depth/Centre/Width (25 x 3 = 75) - this is the raw shape of the spectrum, not mineral names.
- ~44 columns: the mineral classification (Min/Wt/Grp/Error) computed by at least three different internal methods (suffixes uTSAS, sTSAS, sTSAV), each producing its own full result set.
- ~32 columns: quality/tracking metadata (Flags, SNR, recovery_rate) and specific named mineral indices GSWA computes directly (e.g. 'Chlorite w2250', 'White Mica w2200').8.

7. Validating the Signal - Mineral vs Depth (Brief Step 1)

The brief's first how-to step: 'Plot mineral % against depth for one hole by hand - you'll see the rock change as you go down. That's your proof the signal is there.'

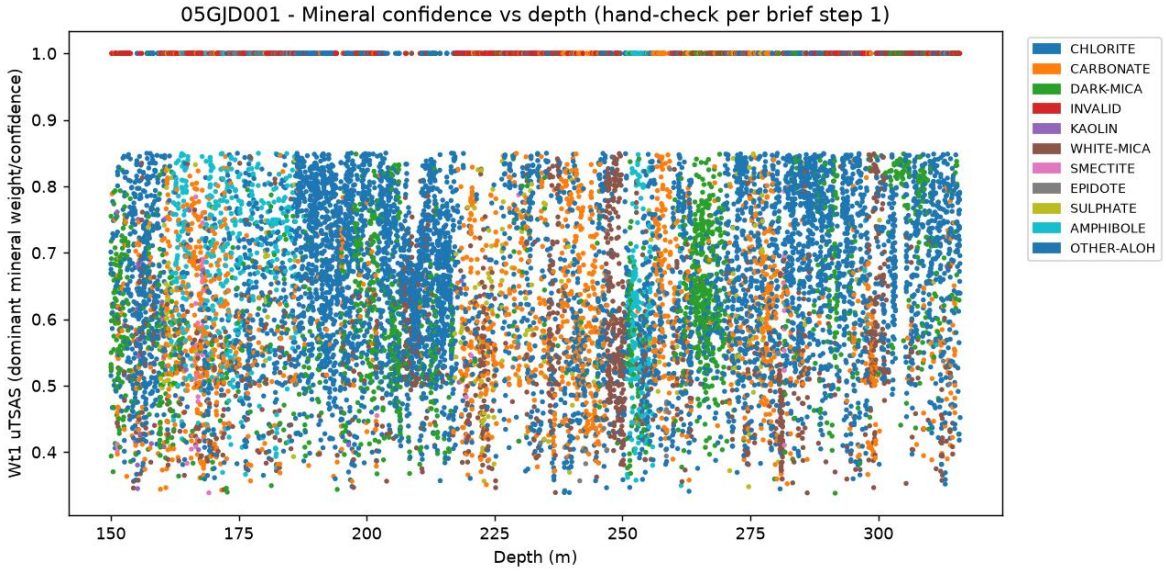
This section is that exercise, done for real against 05GJD001's actual TSA data (Section 7.6).

7.1 Real findings

- Classified depth range: 150.01 m to 315.98 m (166 m of core with at least one mineral identified).

- 11 distinct mineral groups (Grp1 uTSAS) appear across the hole: CHLORITE (6,500 samples), CARBONATE (2,065), DARK-MICA (1,913), INVALID (1,754), WHITE-MICA (1,483), OTHER-ALOH (928), AMPHIBOLE (847), SULPHATE (369), SMECTITE (58), EPIDOTE (50), KAOLIN (39).
- CHLORITE dominates overall, but the mix visibly shifts with depth - e.g. an AMPHIBOLE-heavy zone clusters around 245-255 m, a WHITE-MICA-heavy zone around 235-245 m - not evenly spread through the hole.

Figure 3 - Mineral confidence (Wt1 uTSAS) vs depth, colored by mineral group (Grp1 uTSAS), for all classified samples in 05GJD001:



7.2 What this proves

The colour bands visibly cluster and shift across depth rather than appearing randomly scattered - this is the signal the brief's step 1 is checking for. The rock genuinely changes character as the hole gets deeper, and the TSA data captures that change without needing raw spectral modelling.

7.3 A data-quality observation worth raising

A large number of samples across the entire depth range sit at exactly Wt1 uTSAS = 1.0, forming a flat line at the top of Figure 3 regardless of mineral group. This looks like it could be a default/sentinel value used in a specific edge case (e.g. when only one feature is detected) rather than a genuine 100% confidence result, but this was not confirmed from source code or documentation - worth asking GSWA whether Wt1 = 1.0 should be trusted at face value or filtered out separately from the -3.402823e+38 'no data' sentinel already known about.

8. WAMEX

8.1 How WAMEX search actually works

The main search interface (wamex.dmp.wa.gov.au/Wamex) is built around report metadata, not drill hole names directly:

- A-number / Ten-number (direct report ID)
- Title, Author, Operator
- Report year / date released range
- Report type, Target commodities
- Keywords, plus an 'Advance filter' for refinement
- A spatial/map-based search - search by location on a GIS map instead of by ID

Results give access to full exploration reports (PDF summaries + raw data files - drilling data, geochemistry, sometimes 3D models) across 105,000+ reports

A-number location: Each hole's .tsg file header contains an Executive Summary field (in the [description] section) with the WAMEX Report ID embedded — e.g. 05GJD001's .tsg file shows WAMEX Report ID: A73613. Read as plain text, not through pyTSG's structured API, which doesn't expose this field.